

Few-Shot In-Context Learning for Implicit Semantic Multimodal Content Detection and Interpretation

Xiuxian Wang[✉], Lanjun Wang[✉], *Member, IEEE*, Yuting Su[✉], *Member, IEEE*, Hongshuo Tian[✉],
Guoqing Jin[✉], and An-An Liu[✉], *Senior Member, IEEE*

Abstract—In recent years, the field of explicit semantic multimodal content research makes significant progress. However, research on content with implicit semantics, such as online memes, remains insufficient. Memes often convey implicit semantics through metaphors and may sometimes contain hateful information. To address this issue, researchers propose a task for detecting hateful memes, opening up new avenues for exploring implicit semantics. The hateful meme detection currently faces two main problems: 1) the rapid emergence of meme content makes continuous tracking and detection difficult; 2) current methods often lack interpretability, which limits the understanding and trust in the detection results. To make a better understanding of memes, we analyze the definition of metaphor from social science and identify the three key factors of metaphor: socio-cultural knowledge, metaphorical tenor, and metaphorical representation pattern. According to these key factors, we guide a multimodal large language model (MLLM) to infer the metaphors expressed in memes step by step. Particularly, we propose a hateful meme detection and interpretation framework, which has four modules. We first leverage a multimodal generative search method to obtain socio-cultural knowledge relevant to visual objects of memes. Then, we use socio-cultural knowledge to instruct the MLLM to assess the social-cultural relevance scores between visual objects and textual information, and identify the metaphorical tenor of memes. Meanwhile, we apply a representative interpretation method to provide representative cases of memes and analyze these cases to explore metaphorical representation pattern. Finally, a chain-of-thought prompt is constructed to integrate the output of the above modules, guiding the MLLM to accurately detect and interpret hateful memes. Our method achieves state-of-the-art performance on three hateful meme detection benchmarks and performs better than supervised training models on the hateful meme interpretation benchmark.

Index Terms—Knowledge retrieval, hateful meme detection, hateful meme interpretation, few-shot, in-context learning, large language model.

I. INTRODUCTION

ONLINE memes, as a popular form of communication on social media platforms, convey metaphors with

collective psychology and cultural identity through the combination of text and images, thereby expressing emotions and thoughts [1], [2]. Although memes are often presented in a humorous way, some are used to spread hateful information. These hateful memes can attack individuals or groups based on targeting identity information such as race, gender, and religion. The spread of such memes can not only spark online controversies but also mislead citizens and guide them towards incorrect values. Therefore, it is crucial to develop technologies capable of detecting and understanding hateful memes. However, unlike detecting and understanding explicit semantic content [3], [4], [5], the implicit semantics conveyed by memes are difficult to directly associate through visual and textual representations alone. Accurate interpretation requires a deeper understanding of cultural background and context. Furthermore, with the rapid development of internet culture, memes are emerging and evolving rapidly, which hinders the development of this technology.

To detect hateful memes, previous work has proposed methods that yielded promising performance [6], [7], [8], [9]. These methods identify content with hate metaphors by understanding how visual and textual modalities interact. However, these existing methods mainly focus on performing hateful memes classification, predicting whether a meme incorporates a hateful metaphor, often overlooking an interpretation of the hateful metaphor. To address this gap, [10] introduces a Hateful memes with Reasons Dataset (HatReD), which provides metaphor interpretation annotations for hateful memes in the Facebook Hateful Memes dataset (FHM) [11]. However, this complex annotation process requires significant time investment by professionals, leading to high annotation costs. This undoubtedly limits the generalization ability and application scope of hateful meme interpretation models. Therefore, there is an urgent need to study training-free hateful meme detection and interpretation methods that do not rely on the extensive manually annotated datasets.

To understand the metaphor expressed in memes, we analyze the definition of metaphor to identify key factors in metaphor formation. As stated in *Campbell's "The philosophy of rhetoric"* [12], metaphor is a figure of speech that describes one person, place, or thing in terms of another and its attributes. Structurally, metaphor consists of tenor and vehicle, where the tenor is the person, place, or thing being represented in the metaphor, and the vehicle is what represents the tenor. In memes, the tenor of metaphor is constructed by combining

Received 6 December 2024; revised 20 February 2025; accepted 5 March 2025. Date of publication 13 March 2025; date of current version 8 September 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62425307, Grant U21B2024, and Grant 62202329. This article was recommended by Associate Editor B. Bao. (Xiuxian Wang and Lanjun Wang contributed equally to this work.) (Corresponding author: An-An Liu.)

Xiuxian Wang, Lanjun Wang, Yuting Su, Hongshuo Tian, and An-An Liu are with the School of Electrical and Information Engineering, Tianjin University, Tianjin 300072, China (e-mail: anan0422@gmail.com).

Guoqing Jin is with the State Key Laboratory of Communication Content Cognition, People's Daily Online, Beijing 100733, China (e-mail: jinguoqing@people.cn).

Digital Object Identifier 10.1109/TCSVT.2025.3550900

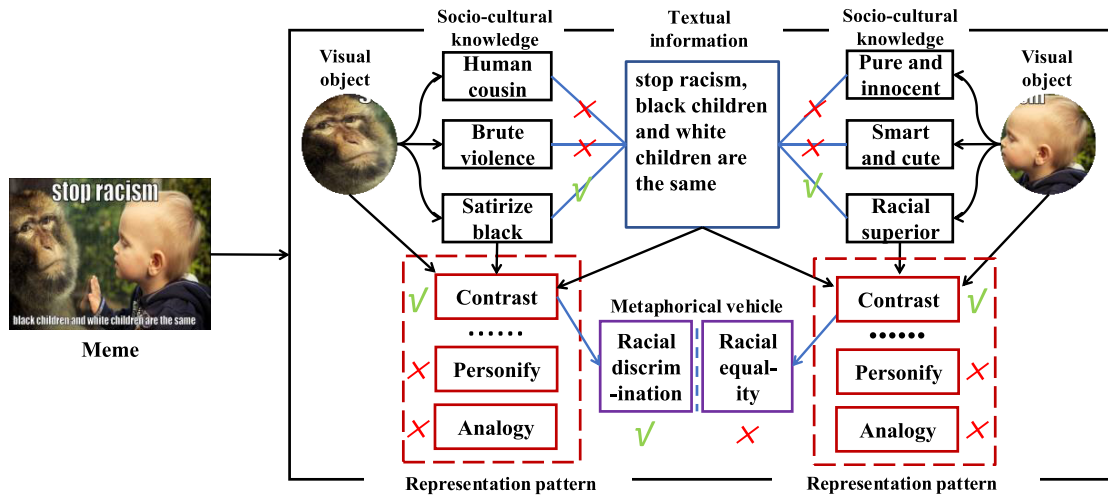


Fig. 1. The process of metaphor formation in memes based on these three key conditions. Firstly, the visual object “chimpanzee” is linked with the text content “stop racism, black children and white children are the same” through sociocultural knowledge “sarcastic about black people”, forming the metaphorical tenor. Secondly, the metaphorical tenor is mapped to the metaphorical vehicle “racial discrimination” through a metaphor expression pattern “conceptual contrast”. Finally, the metaphorical vehicle “racial discrimination” is used to express hateful emotion towards black people.

visual objects with textual information. This combination process is based on socio-cultural knowledge of visual object to associate the visual objects with the textual information. For example, in Figure 1, “chimpanzee” is the visual object, “stop racism / black children and white children are the same” is the textual information, and “satirizing black” is the socio-cultural knowledge of the “chimpanzee”. The “chimpanzee” is linked to the textual information based on “satirizing black.” Furthermore, there are representation patterns in the metaphor, e.g., contrast, personify, analogy, etc [13]. The tenor can be represented as a vehicle by a representation pattern. In Figure 1, the vehicle “racial discrimination” is represented from the tenor through the representation pattern “contrast” thus conveying a negative sentiment toward black individuals. Based on this analysis, three key factors of the metaphor formation in memes: 1) metaphorical tenor; 2) socio-cultural knowledge; and 3) metaphorical representation pattern.

While investigating the three key factors of metaphor, we encounter three significant challenges: 1) It is not a trivial task to identify the visual object in the tenor based only on textual information. Memes often contain multiple visual objects, but not all visual objects are socio-culturally associated with the textual information. 2) Socio-cultural knowledge (e.g., racial superior in Fig. 1) is a bridge to construct the tenor based on the visual object and text information. However, understanding and recognizing this specific knowledge is based on personal experiences [14], [15]. In addition, socio-culture knowledge varies in different geographic and demographic regions and sometimes is emerging and changing rapidly. 3) Identifying metaphorical representation patterns is also a challenge. A straightforward method is to analyze and summarize similar cases for understanding these patterns. However, common retrieval methods [16], [17], [18], [19], [20] often retrieve similar cases based on some widely used feature extraction

methods (e.g., CLIP [21]) which do not consider implicit semantics.

In this paper, we propose a few-shot in-context learning multimodal hateful meme detection and interpretation framework by exploring the key factors of metaphor, providing an out-of-the-box solution for multimodal hateful meme detection and interpretation. Our framework addresses the above challenges through the following steps: 1) Socio-Cultural knowledge prompt-driven multimodal Generative Search (SCGen Search) module. This module first constructs socio-cultural knowledge prompt from five dimensions: race, nationality, gender, religion, and disability. The socio-cultural knowledge prompt guide a Large Language Model (LLM) to search socio-cultural knowledge related to visual objects. Then, the LLM summarizes the search results and generates socio-cultural knowledge assertions about the visual objects. 2) Socio-Cultural Relevance Assessment-based Metaphorical Tenor Identification (SCRA-MTI) module. This module first establishes a socio-cultural relevance scoring level, which guides the LLM based on socio-cultural knowledge assertions to assess the relevance score between visual objects and textual information. Subsequently, the visual objects are ranked according to this relevance score, forming a candidate set of visual objects that constitute the metaphorical tenor. 3) Representative Case Retrieval (RCR) module. This module aims to find representative cases with representative interpretation by analyzing the common decision-making logic of a large number of similar cases in the hateful meme detection model. The task adaptability and explainability of these cases can guide the model to identify metaphorical representation pattern. Finally, inspired by the process of forming metaphors depicted in Figure 1, we construct a chain of thought for interpreting metaphor. This chain of thought guides the MLLM to execute hateful meme detection and interpretation by integrating the outputs of the aforementioned three steps.

We summarize our contributions as follows:

- To understand the metaphor expressed in memes, we discover three key factors of metaphor through analyze the definition of metaphor: 1) metaphorical tenor; 2) socio-cultural knowledge; and 3) metaphorical representation pattern.
- We propose a novel few-shot in-context learning multimodal hateful meme detection and interpretation framework by exploring the key factors of metaphor, providing an out-of-the-box solution for multimodal hateful meme detection and interpretation.
- Experimental results demonstrate that our approach can achieve performance surpassing fully supervised methods, even without training or fine-tuning on extensive manually annotated datasets.

II. RELATED WORKS

A. Hateful Meme Detection

In recent years, hateful meme detection is an emerging multimodal task that aims to detect memes containing hateful content. Existing researches investigate the classic two-stream models that integrate text and visual features for classifying hateful memes [11], [22]. Additionally, studies fine-tune large-scale pre-trained multimodal models to adapt to the multimodal classification tasks [6], [7], [8], [9], [23], [24]. While these studies make some progress in classification tasks, they generally lack the interpretation. To improve the interpretation of hateful meme detection methods, [25] proposes a post-hoc explanation framework to examine the visual-text slur grounding learned by pre-trained multimodal models trained to perform the hateful meme classification task. However, the framework still falls short in providing informative reasons to explain hateful memes. Recent research introduces a hateful memes with reasons dataset, which provides metaphorical interpretation annotations for hateful memes. However, the high cost of annotating metaphorical explanations limits the scale of dataset. This undoubtedly limits the generalization ability. Therefore, we propose a few-shot in-context learning multimodal hateful meme detection and interpretation framework by exploring the key factors of metaphor formation. Specifically, this study first clarifies the three key factors of metaphor formation through analysing the definition of metaphor. Subsequently, we design three critical steps for accurately extracting these key factors from memes. Finally, we use these three key factors to guide Large Language Model (LLM) [26], [27] to analysis the metaphors within memes step-by-step. Through this approach, we can more effectively understand the metaphorical meanings in memes, thereby enabling accurate detection and interpretation of hateful memes.

B. Large Language Models and Multimodal Large Language Models

In the field of natural language processing, large language models (LLMs) achieve remarkable progress. Based on the Transformer [28] architecture, LLMs learn rich language knowledge and semantic representations through unsupervised

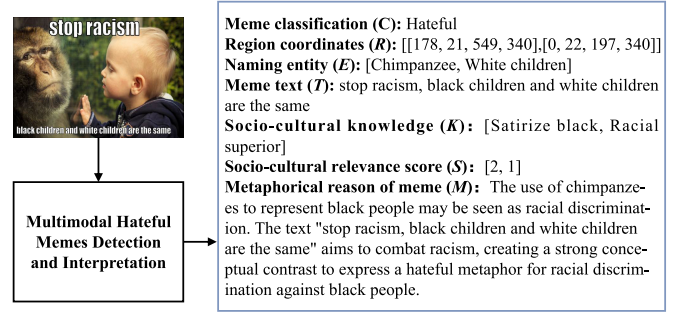


Fig. 2. The task of multimodal hateful meme detection and interpretation generation. Given a meme image I , the goal of the multimodal hateful meme detection and interpretation task is to generate a meme classification C and a structured interpretation $\{R, E, T, K, S, M\}$, where R is the set of region coordinates of visual objects, E is the set of named entity of visual objects, T is the textual information, K is the set of socio-cultural knowledge of visual objects, S is the set of socio-cultural relevance score between the visual objects and the textual information, and M is the metaphorical reason of the meme.

pre-training on large-scale text data. Models such as GPT-3 [29], PaLM [30], LLaMA [31], and GPT-4 [32] can generate fluent and natural text and complete various natural language processing tasks, including text generation, question-answering systems, and machine translation. These models learn the statistical relationships and semantic associations between words in a large amount of text, thus possessing powerful language understanding and generation capabilities. In practical applications, they significantly improve the efficiency and quality of text processing, bringing new solutions to fields such as intelligent customer service and content creation.

However, LLMs mainly focus on text data and have limitations in processing non-text information such as images and audio. To overcome this limitation, multimodal large language models (MLLMs) emerge. MLLMs aim to integrate text with data from other modalities (such as images, audio, video, etc.) to achieve more comprehensive information understanding and interaction. For example, models like LLaVA [33], MiniGPT-4 [34], and Gemini [35] can perform tasks such as image caption generation and image question-answering by aligning and fusing image features with text features. By introducing techniques such as cross-modal attention mechanisms, MLLMs enable the model to learn the associations between different modal data, expanding the application boundaries of large language models. This allows them to handle more complex and diverse tasks.

III. OUR APPROACH

As shown in Figure 2, given a meme image I , the goal of the multimodal hateful meme detection and interpretation task is to generate a meme classification C and a structured interpretation $\{R, E, T, K, S, M\}$, where R is the set of region coordinates of visual objects, E is the set of named entity of visual objects, T is the textual information, K is the set of socio-cultural knowledge of visual objects, S is the set of socio-cultural relevance score between the visual objects and the textual information, and M is the metaphorical reason of the meme.

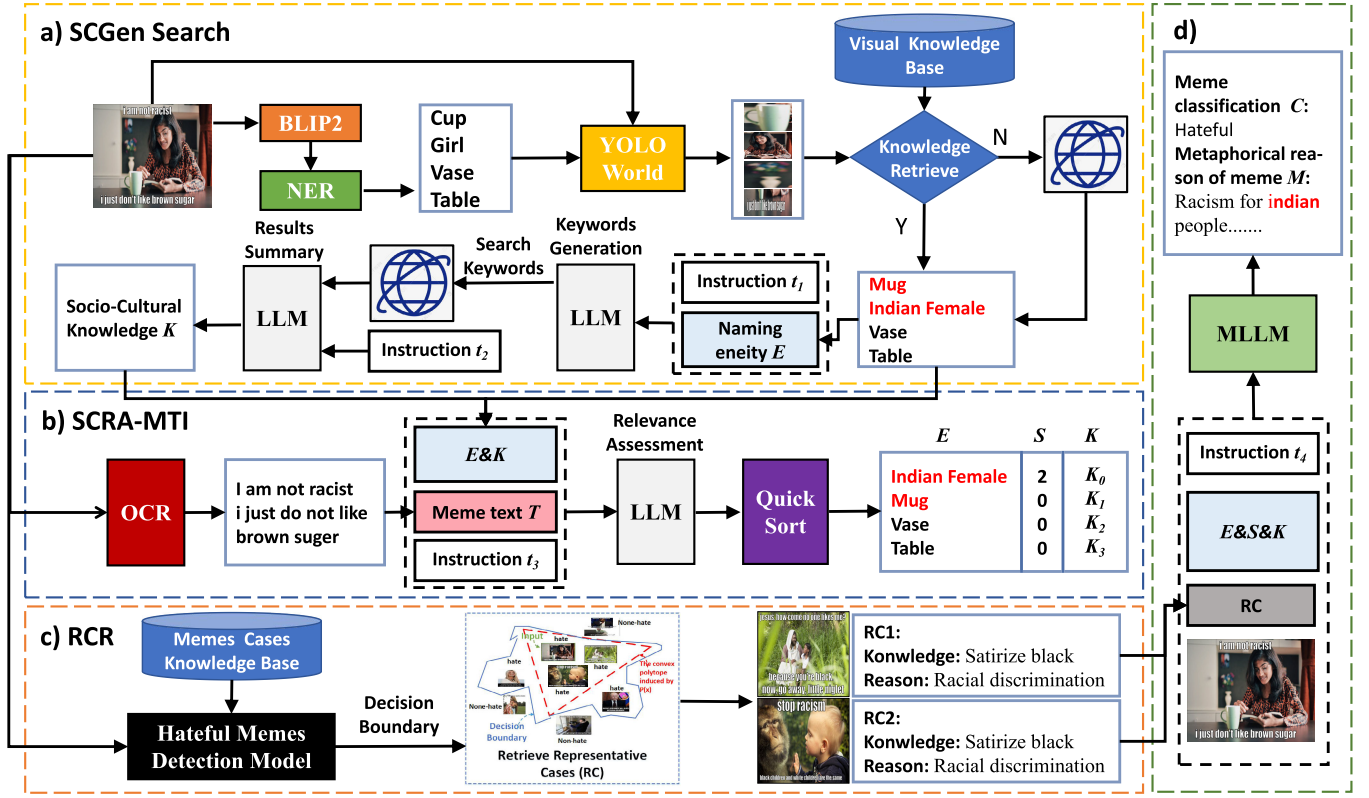


Fig. 3. The architecture of the few-shot multimodal hateful meme detection and interpretation framework. a) Socio-Cultural knowledge prompt-driven multimodal Generative Search (SCGen Search) module. b) Socio-Cultural Relevance Assessment-based Metaphorical Tensor Identification (SCRA-MTI) module. c) Representative Case Retrieval (RCR) module. d) hateful meme detection and interpretation generation.

To address this task, we propose a few-shot multimodal hateful meme detection and interpretation framework by exploring the key factors of metaphor, as shown in Figure 3. Our method consists of four steps: 1) Socio-Cultural knowledge prompt-driven multimodal Generative Search (SCGen Search) module, which aims to search socio-cultural knowledge related to visual objects in meme. 2) Socio-Cultural Relevance Assessment-based Metaphorical Tensor Identification (SCRA-MTI) module, which aims to assess the socio-cultural relevance score between visual objects and textual information based on socio-cultural knowledge. It then ranks them to obtain a candidate set of visual objects that constitute the metaphorical tenor, in accordance with the relevance score. 3) Representative Case Retrieval (RCR) module, which aims to find a set of cases with representative interpretations for the input meme, named representative cases, to guide the model in learning the metaphorical representation pattern. Finally, referring to the metaphor formation process in Figure 1, a chain of thought is constructed to guide the MLLM to integrate the outputs of the previous three steps for the detection and interpretation of hateful memes.

A. Socio-Cultural Knowledge Prompt-Driven Multimodal Generative Search

The metaphorical tenor is constructed by utilizing socio-cultural knowledge of visual object to link visual object with textual information. Therefore, acquiring the socio-cultural knowledge of visual objects is a necessary step for identifying

the metaphorical tenor in meme. To achieve this goal, the first task is to extract visual objects from the memes. By accurately identifying and extracting these visual objects, we can provide clear targets for subsequent knowledge search. Next, it is necessary to determine the information about the named entity of these visual objects, such as people, places, or events. This step not only narrows the scope of knowledge search but also provides accurate keywords for the search. Finally, we need to obtain rich and accurate socio-cultural knowledge. Through generative search, we can filter out highly relevant knowledge related to visual objects from massive amounts of online information. Based on the above analysis, we propose an SCGen Search module, which consists of three key steps: 1) Extracting visual objects; 2) Retrieving naming entities for visual objects; 3) Obtaining relevant socio-cultural knowledge through generative search. The detail steps are as follows.

1) *Extracting Visual Objects*: When processing meme images, the openness of visual scenes and the non-fixity of visual object categories make it difficult for generic object detection models to be directly applied. To accurately extract visual objects from memes, we adopt a comprehensive approach that combines image captioning, named entity extraction, and open-vocabulary object detection. The specific steps are as follows:

We first use the image captioning model BLIP2 [36] to generate an image caption IC for the meme image I . Subsequently, we use a large language model Qwen2 [37] to extract a named entity from IC as a set of category

keywords $CK = \{ck_i\}^{N_{ck}}$, where ck_i represents the i -th category keyword, and N_{ck} is the total number of category keywords. Finally, we use the category keywords to guide the open vocabulary object detection model YOLO-World [38] to detect visual objects in the meme, obtaining the set of region coordinates for visual objects $R = \{r_i\}^{N_r}$, where r_i represents the i -th region coordinate of visual object, and N_r is the total number of visual objects.

$$\begin{aligned} IC &= \text{BLIP2}(I) \\ CK &= \text{Qwen2}(IC) \\ R &= \text{YOLO-World}(CK) \end{aligned} \quad (1)$$

2) *Retrieving Naming Entities for Visual Objects*: Identifying the specific named entity information of the extracted visual objects not only narrows the scope of knowledge search but also provides accurate keywords for the search. To achieve this goal, we use existing datasets of hateful meme detection to construct a visual knowledge base, which stores visual feature vectors along with their corresponding named entity information. Furthermore, to overcome the limitations of local knowledge bases in terms of knowledge volume, we use Baidu Image Recognition API to search for visual objects that could not be retrieved with relevant information about the named entity from the local knowledge base. The specific steps are as follows:

Firstly, a visual socio-cultural knowledge base is constructed based on existing open-source hateful meme detection datasets [11], [39], [40]. The knowledge base stores visual feature vectors $F_k = \{f_{kj}\}^{N_k}$ and their corresponding named entities $E_k = \{e_{kj}\}^{N_k}$, where f_{kj} , e_{kj} is the j -th visual feature vector and named entity. Reference [41] demonstrates the impressive image-matching ability of DINOv2 [42], which robustly captures similar semantic intent across different objects and even distinct images. Therefore, following the experimental setup described in [41], we employ DINOv2 [42] to extract visual feature vectors, where the dimensionality of the visual feature vectors is 768. Secondly, according to the coordinates of the region of the visual objects R , we use the DINOv2 to extract the feature vectors of the visual objects $F = \{f_i\}^{N_r}$, where f_i is the i -th feature vector of the visual object. Thirdly, we perform visual similarity matching by computing the L2 Euclidean distance between f_i and f_{kj} . If the L2 distance is less than or equal to a predefined threshold α , the match is considered successful, and e_{kj} is assigned to the corresponding visual object. For visual objects that do not achieve a successful match in the knowledge base, we utilize the Google image recognition API to search the corresponding named entities on the Web. Finally, we obtain a set of named entities for visual objects $E = \{e_i\}^{N_r}$, where e_i is the i -th named entity of visual object.

$$F = \text{DINOv2}(R) \quad (2)$$

$$e_i = \begin{cases} e_{kj} & L2(f_i, f_{kj}) \leq \alpha \\ \text{API}(f_i) & L2(f_i, f_{kj}) > \alpha \end{cases} \quad (3)$$

where the threshold α is determined by calculating the average of the L2 distances between correctly matched samples.

0 (Unrelated): There is no obvious socio-cultural connection between the two contents. They may come from completely different cultural backgrounds, and the topics or values discussed have no intersection.

1 (Relevant): There is a clear connection between the two contents at the socio-cultural level, but this connection is not direct or profound.

2 (Strongly relevant): The two contents have a direct and profound connection at the socio-cultural level, and they may jointly explore a specific social issue, historical event, or cultural phenomenon, with many intersections and interactions between them.

Fig. 4. The socio-cultural relevance scoring level. Unrelated is 0 score; Relevant is 1 score; Strongly relevant is 2 score.

3) *Search for Socio-Cultural Knowledge Related to Visual Objects*: In searching for socio-cultural knowledge related to visual objects, it is crucial to filter out highly relevant information from the vast amount of online data, ensuring the accuracy and relevance of the search results. To achieve this objective, we first construct a socio-cultural knowledge prompt for each named entity e_i from five perspectives: ethnicity, nationality, gender, religion, and disability [43]. This prompt guides the LLM to generate search keywords SK for searching socio-cultural knowledge. The socio-cultural knowledge prompt consists of instruction and the named entity, the instruction as follow: “Analyze the socio-cultural characteristic attributes of {named entity} from five angles: ethnicity, nationality, gender, religion, and disability, and generate some search keywords for searching socio-cultural knowledge about these characteristic attributes.” Formally, we denote the instruction as t_1 . The socio-cultural knowledge prompt p_{ki} can be defined as:

$$p_{ki} = [t_1, e_i] \quad (4)$$

$$SK = \text{LLM}(p_{ki}) \quad (5)$$

where $[\cdot; \cdot]$ denotes text concatenation.

Subsequently, according to the search keywords SK , the google web search API is used to search for relevant socio-cultural knowledge on the internet, and design a instruction t_2 to guide the LLM to summarize the search results SR , generating the socio-cultural knowledge assertions related to the named entity. The instruction t_2 as follow: “Please summarize the {search results} of socio-cultural knowledge about {named entity} found online, avoiding irrelevant content.” Finally, we acquire a set of socio-cultural knowledge assertions of visual objects $K = \{k_i\}^{N_r}$.

$$\begin{aligned} SR &= \text{API}(SK) \\ k_i &= \text{LLM}(SR) \end{aligned} \quad (6)$$

B. Socio-Cultural Relevance Assessment-Based Metaphorical Tenor Identification

After acquiring socio-cultural knowledge related to visual objects, effectively leveraging this knowledge to establish connections between visual objects and textual information is a critical challenge in identifying the metaphorical tenor. To address this challenge, we use socio-cultural knowledge

of the visual object as a reference and leverage the robust reasoning capabilities of LLM to assess the socio-cultural relevance between visual objects and textual information. The specific steps are as follows:

Firstly, we design a socio-cultural relevance scoring level L to assess the socio-cultural relevance between visual objects and textual information, as shown in Figure 4. Subsequently, combining the socio-cultural knowledge assertions of the i -th visual object k_i , the textual information T , and the socio-cultural scoring level L , we construct a socio-cultural relevance assessment prompt p_{ri} for i -th named entity e_i to guide LLM to assess the socio-cultural relevance score between e_i and T . The instruction of prompt is as follows: “According to the {socio-cultural relevance scoring level}, please assess the social-cultural relevance score between {named entity} and {textual information} based on {the social-cultural knowledge of the named entity}.” Finally, based on the socio-cultural relevance scores $S = \{s_i\}^{N_r}$, we sort the named entities to obtain a candidate set of the metaphorical tenor $Z = \{z_i\}^{N_r}$ according to the relevance score, where z_i is the named entity with the i -th highest socio-cultural relevance score. The above operator can be formally defined as follows:

$$\begin{aligned} p_{ri} &= [t_3, e_i, T, L] \\ s_i &= \text{LLM}(p_{ri}) \\ S &= \{s_0, s_1, \dots, s_{N_r}\} \\ Z &= \text{quick_sort}(S, E) \end{aligned} \quad (7)$$

where t_3 is the instruction of socio-cultural relevance assessment prompt, s_i is the i -th socio-cultural relevance score of named entity, quick_sort is the quick sort algorithm [44].

C. Representative Case Retrieval

In the “*Oxford English Dictionary*” [13], the meme is defined as “a basic unit of culture that is transmitted by non-genetic means, especially imitation.” This definition underscores imitation is the primary mechanism for meme dissemination. Therefore, studying the metaphorical representation pattern within memes by referring to similar cases is a feasible strategy. However, existing Retrieval Augmented Generation (RAG) systems typically employ knowledge retrieval methods that can only retrieve knowledge cases based on generic visual and textual features [45]. These methods exhibit limited adaptability and interpretability in hateful meme detection and interpretation task. To mitigate this limitation, we leverage a representative case retrieval method based on representative interpretation [46]. Our method identifies cases with representative interpretations for input meme by leveraging the common decision logic of a large number of similar cases in a hateful meme detection model. The adaptability and interpretability of these representative cases guide the model in exploring the metaphorical representation pattern in memes. The specific steps are outlined as follows.

Firstly, we use the human-annotated hateful memes with reasons dataset as memes cases knowledge base B_{ck} . Then, we employ the representative interpretation method to find representative cases $RC = \{rc_k\}^{N_{rc}}$ of input meme I from

TABLE I
STATISTICAL DISTRIBUTIONS OF DATASETS USED FOR EVALUATION

Datasets	Train		Test	
	Hate	None-hate	Hate	None-hate
FHM [11]	3050	5450	250	250
HarMeme [39]	1064	1949	124	230
MAMI [40]	5000	5000	500	500

B_{ck} , where rc_k is the k -th representative case of input meme I , N_{rc} is the number of representative cases of input meme I . Since these similar memes share common decision logic in the hateful meme detection (HMD) model [47], their have category relevance and potential associations of metaphor, facilitating model learning and understanding. The above operator can be formally defined as follows:

$$RC = RI(\text{HMD}, I, B_{ck}) \quad (8)$$

where RI is the representative interpretation method [46].

D. Hateful Meme Detection and Interpretation

To address the issue of neglecting important information when directly feeding knowledge into MLLM, we construct a Chain of Thought (CoT) [48] by simulating the metaphor formation process shown in Figure 1. We use the CoT to guide the MLLM to integrate the output of three key steps to operate hateful meme detection and generate metaphorical reasons. The instruction of CoT is as follows: Firstly, analyze the relevance between the {meme image} and {representative cases} based on {metaphorical tenors}, {socio-cultural knowledge}, {textual information}, and {socio-cultural relevance score}. Secondly, explore the metaphorical representation pattern from the representative cases. Finally, combine the aforementioned information for hateful meme detection and metaphor analysis, generating the meme classification: {0 non-hate/1 hate} and structured interpretations: {structured interpretations}. The above operator can be formally defined as follows:

$$\text{CoT} = [t_4, I, RC, Z, K, T, S] \quad (9)$$

$$C, \{R, E, T, K, S, M\} = \text{MLLM}(\text{CoT}) \quad (10)$$

where C is the binary label and $\{R, E, T, K, S, M\}$ is a structured interpretation as shown in Figure 2, t_4 is the instruction of CoT.

IV. EXPERIMENT

In this section, we first introduce the experimental setup, including implementation details, data and processing, and the selection of evaluation metrics. Then, to validate the effectiveness of our method in the task of hateful meme detection, we conduct comparative experiments with state-of-the-art baseline models across three mainstream datasets, supplemented by ablation studies. Lastly, to further verify the relevance and fluency of the metaphorical reasons generated by our method related to memes, we carry out comparative experiments and case studies on the HatReD dataset.

TABLE II
HATEFUL MEME DETECTION PERFORMANCE COMPARISONS ON FHM, MAMI AND HarM DATASET

Model	Training	FHM		MAMI		HarM	
		AUC	Acc	AUC	Acc	AUC	Acc
Text BERT [49]	✓	66.10	57.12	74.48	67.37	81.39	75.68
Image-Region [11]	✓	56.69	52.34	70.20	64.18	76.46	73.05
CLIP-BERT [39]	✓	66.97	58.28	77.66	68.44	82.63	80.48
DisMultiHate [23]	✓	69.11	62.42	78.21	70.58	83.69	78.05
VisualBERT COCO [50]	✓	68.71	61.48	78.71	71.06	80.46	75.31
ViLBERT CC [51]	✓	73.05	64.70	77.71	69.48	84.11	78.70
MMBT-Region [52]	✓	72.86	65.06	79.17	70.46	85.48	79.83
PromptHate [8]	✓	76.76	67.82	76.21	68.08	87.51	79.38
BLIP [53]	✓	76.80	69.20	80.59	71.84	87.09	81.81
ALBEF [54]	✓	79.40	70.58	83.24	72.77	85.49	80.99
Pro-CapBERT [55]	✓	77.50	68.14	79.62	71.06	89.04	82.06
Pro-CapPromptHate [55]	✓	80.87	72.28	82.53	73.06	90.25	83.25
LLaVA (Vicuna-13B) [56]	✓	85.30	77.30	-	-	-	-
Spark-VL	✗	67.80	58.65	73.41	66.40	71.40	69.60
Qwen-VL-Max	✗	65.20	55.60	74.80	65.90	73.46	67.31
GPT4	✗	78.20	68.80	77.20	67.50	77.09	73.06
Spark-VL w/ Ours	✗	82.60	73.20	81.10	72.18	86.11	80.08
Qwen-VL-Max w/ Ours	✗	80.60	72.80	83.34	73.40	85.46	79.20
GPT4 w/ Ours	✗	87.60	78.60	84.30	74.63	90.51	83.29

TABLE III
AUTOMATIC EVALUATION RESULTS OF THE GENERATED REASONS ON HATRED TEST SET

Model	Training	N-gram matching	
		BLEU	ROUGE-L
VisualBERT-GPT2 [10]	✓	0.065	0.219
VisualBERT-RoBERTa [10]	✓	0.179	0.391
VL-T5 [59]	✓	0.180	0.378
GPT4	✗	0.122	0.333
GPT4 w/ Ours	✗	0.217	0.429

TABLE IV
HUMAN EVALUATION RESULTS OF THE GENERATED REASONS ON HATRED TEST SET

Model	Training	Metrics (Avg.)	
		Fluency	Relevance
VisualBERT-GPT2 [10]	✓	4.870	2.283
VisualBERT-RoBERTa [10]	✓	4.344	2.931
VL-T5 [59]	✓	4.626	2.602
GPT4	✗	4.951	3.110
GPT4 w/ Ours	✗	4.957	3.748

A. Implementation Details

In the process of retrieving representative cases, the hateful meme detection model used is a ResNet50 [58] model trained on FHM, MAMI, and HarM datasets. The computing resources for training the hateful meme detection model are configured as follows: the GPU is one NVIDIA GeForce RTX 4090; the CPU is one AMD EPYC 9754 128-Core Processor; the memory capacity is 60GB; and the hard-disk

space is 50GB. During the stages of generative search and socio-cultural relevance assessment, we employ ChatGPT as large language models, with the temperature parameter set to 0.6 and the maximum generation length set to 512. In the phase of hateful meme detection and interpretation generation, we conduct extensive testing on the multimodal large language models GPT-4, Qwen-VL-Max, and Spark-VL. The computing resources for our framework are the same as those for training the hateful meme detection model.

B. Dataset and Preprocessing

We evaluate the hateful meme detection performance of our method on three datasets: Facebook Hateful Memes [11], HarMeme [39], and MAMI [40]. We evaluate the meme reasons generation performance of our method on HatReD [10] dataset. The detailed information of the datasets and the corresponding preprocessing method (if applied) are as follows. The statistical distribution of the datasets used for evaluation is shown in Table I.

1) *Facebook Hateful Memes Dataset*: The Facebook Hateful Meme (FHM) dataset is constructed by facebook. It contains synthetic memes with added confounding factors, i.e., no representative hateful metaphors in either single-modal visual features or textual features. Therefore, unimodal information is not sufficient for detection, and deep multimodal reasoning is required. The FHM dataset contains hateful meme targeting various vulnerable groups, such as religion, race, gender, nationality, and disability.

2) *HarMeme Dataset*: The HarMeme dataset consists of COVID-19 related memes collected from twitter, which are labeled with three classes: very hateful, partially hateful, and

non-hateful. In line with previous works [55], we combine the very hateful and partially hateful memes into hateful memes.

3) *MAMI Dataset*: The MAMI dataset is dedicated to the study of a specific type of hateful meme, namely those that target women as their victims. Therefore, the performance on the MAMI dataset can demonstrate the effectiveness and capability of hateful meme detection techniques in addressing female victims.

4) *HatReD Dataset*: Hateful meme with Reasons Dataset (HatReD), which is a new multimodal hateful meme dataset annotated with the underlying hateful contextual reasons. Based on the HatReD, researchers define a new conditional generation task that aims to automatically generate underlying reasons to explain hateful meme.

C. Evaluation Metrics

1) *Hateful Meme Detection*: To ensure fairness in the comparison, we adopt the evaluation metrics commonly used in current hateful meme detection researches [11], [55]. Accuracy (Acc) and Area Under the receiver operating characteristic Curve (AUC) [59]. In this paper, to test the AUC metric of the method based on MLLM, we guide MLLM to generate metaphor category probabilities for the purpose of evaluating model performance.

2) *Hateful Memes Reason Generation Task*: To ensure fairness in the comparison, we adopt the evaluation metrics commonly used in current hateful meme reason generation research [10]. This process involves two evaluation methods: automatic metric assessment and human evaluation. The automatic evaluation adopt metrics commonly used in natural language generation tasks, such as BLEU [60] and ROUGE-L [61]. Based on these metrics, we can quantify the proximity between the generated metaphorical reason and the ground truth. For human evaluation, we recruit human evaluators to assess the generated reasons on two aspects: fluency and relevance. The human evaluators are tasked to rate the generated reasons on the Likert scales [62].

D. Results of Hateful Meme Detection

To validate the effectiveness and robustness of our proposed method, we compare it with two types of methods on three mainstream datasets for hateful meme detection. 1) Supervised learning methods: They train models from scratch on hateful meme detection data in a fully supervised paradigm, which includes most of the state-of-the-art methods. 2) Zero-shot baselines: Unlike supervised methods, our approach does not require training on hateful meme detection data. Due to the difficulty, there is a lack of zero-shot methods for hateful meme detection in existing works. Therefore, we test the zero-shot performances of several multimodal large language models (MLLM) on the hateful meme detection task as our baselines, including GPT4 [7], Qwen-VL-Max [9], and Spark-VL. From the results in Table II, we have two key observations: 1) Despite not being fine-tuned or trained on the hateful meme detection dataset, our method can still achieve performance that surpasses the supervised learning methods. This validates that our idea of exploring the key

factors of metaphors formation to understand memes is a effective approach. 2) Our approach outperforms the zero-shot baselines that solely utilize a single MLLM, such as GPT4 and Qwen-VL-Max. The performance improvement indicates that supplementing inputs with sociocultural knowledge can effectively construct multimodal information associations, and exploring metaphor representation patterns from representative cases can guide MLLM to accurately comprehend the metaphorical information mapped by multimodal information associations.

E. Results of Hateful Memes Reason Generation Task

In this section, our objective is to validate the capability of our model to generate fluent and relevant reasons for hateful memes. This process involves two evaluation methods: automatic metric assessment and human evaluation.

1) *Automatic Evaluation*: Table III shows the automatic evaluation of the generated reason for hateful memes in HatReD. We observe that our approach outperforms both fine-tuned vision language models and baseline model GPT4. Notably, despite not relying on any additional training datasets, our method surpasses the fine-tuned models in terms of relevance to the ground truth. This improvement can be attributed to several key factors: 1) The integration of socio-cultural knowledge provides clear guidance, resulting in more precise content in terms of topic and background knowledge; 2) By introducing representative cases as exemplars for model learning, these instances not only offer content-based examples but also demonstrate specific expressions of the ground truth, guiding the model to produce outputs that are structurally and stylistically closer to the ground truth. The application of this comprehensive approach significantly enhances the quality of the generated metaphorical reason for memes.

2) *Human Evaluation*: As shown in Table IV, The human evaluation results further corroborate the findings from the automatic evaluation. Our method not only achieved higher fluency scores but also significantly outperformed fine-tuned models in terms of relevance. Due to their strong language understanding and generation capabilities, large-scale language models such as GPT-4 have achieved a high level of logical coherence and organization in the text they produce. Consequently, these models demonstrate significant advantages in terms of text fluency, with relatively limited room for further improvement. However, relevance, as an important metric for evaluating the alignment between generated content and target tasks (e.g., explaining the rationale of memes), still holds considerable potential for optimization. Experimental results show that our method achieves a substantial improvement in the relevance metric compared to baseline models, with a performance gain of 20.5%. This outcome highlights two principal strengths of our approach: 1) Maintaining output quality without reliance on specific training datasets; 2) Ensuring naturalness and acceptability of the output while accurately capturing metaphorical expressions related to the meme.

TABLE V
ABLATION STUDY

Model	FHM		MAMI		HarM	
	AUC	Acc	AUC	Acc	AUC	Acc
Spark-VL	67.80	58.65	73.41	66.40	71.40	69.60
Qwen-VL-Max	65.20	55.60	74.80	65.90	73.46	67.31
GPT4	78.20	68.80	77.20	67.50	77.09	73.06
Spark-VL + SCK	73.50	65.80	78.00	68.50	81.38	74.00
Qwen-VL-Max + SCK	76.50	67.20	77.24	69.08	78.62	73.50
GPT4 + SCK	81.40	73.10	80.53	71.06	84.90	78.52
Spark-VL + SCK + SCRS	79.10	69.40	79.37	70.48	83.01	76.50
Qwen-VL-Max + SCK + SCRS	80.40	68.95	81.00	71.40	82.63	76.68
GPT4 + SCK + SCRS	83.76	74.52	83.19	71.94	86.49	80.31
Spark-VL + SCK + SCRS + RC	82.60	73.20	81.10	72.18	86.11	80.08
Qwen-VL-Max + SCK + SCRS + RC	80.60	72.80	83.34	73.40	85.46	79.20
GPT4 + SCK + SCRS + RC	87.60	78.60	84.30	74.63	90.51	83.29

TABLE VI
AUTOMATIC EVALUATION RESULTS OF THE GENERATED REASONS ON HATRED TEST SET

Model	Complexity	Inference Time
GPT4	O(n)	3.00s
GPT4+SCK	O(n)	6.71s
GPT4+SCK+SCRS	O(n)	7.94s
GPT4+SCK+SCRS+RC	O(n)	8.05s

F. Ablation Study

1) Effectiveness of Introducing Socio-Cultural Knowledge:

To investigate the effects of introducing socio-cultural knowledge on the performance of hateful meme detection, we conduct ablation study on three different baseline models (Spark-VL, Qwen-VL-Max and GPT4). The experiments are performed under two experiment settings: 1) As shown in rows 1-3 of Table V, Spark-VL, Qwen-VL-Max and GPT4 are the experiment setting 1; 2) As shown in rows 4-6 of Table V, Spark-VL + SCK, Qwen-VL-Max + SCK and GPT4 + SCK are the experiment setting 2. Table V presents the performance comparison of the baseline models under the two experimental setups, where the SCK represent socio-cultural knowledge. By comparing these two sets of experiment, we observe that the performance of three baseline models improve on all three datasets. Notably, on the FHM dataset, the AUC and Acc performance of Qwen-VL-Max improved by 11.3% and 11.6%, respectively. These findings indicate that introducing socio-cultural knowledge has a positive effect on enhancing the performance of hateful meme detection. More specifically, introducing socio-cultural knowledge improves the ability of model to understand the socio-cultural context within memes, thereby effectively identify the memes containing hateful content.

2) Effectiveness of Incorporating Socio-Cultural Relevance Score:

To evaluate the effects of introducing Social-Cultural Relevance Scores (SCRS) on model performance, we design a new experiment setting 3, as shown in rows 7-9 of Table V. By comparing the results from the experiment setting 3 with

the experiment setting 2, we observe that the performance of three baseline models improve on all three datasets. This result suggests that by prioritizing visually and textually informative content with high social-cultural relevance scores, the model effectively mitigates the noise introduced by less relevant information, thereby enhancing overall performance. The introduction of SCRS enables the model to focus more accurately on salient features within the data, which is critical for improving the accuracy of its classifications.

3) Effectiveness of Incorporating Representative Cases:

To further investigate the effects of introducing representative cases in enhancing model performance, we design experiment setting 4 as a comparative analysis against experiment setting 3. In experiment setting 4, we incorporate Representative Cases (RC) into the experimental setup that is uses in experiment setting 3. As shown in rows 10-12 of Table V, The experiment setting 4 as follow: Spark-VL + SCK + SCRS + RC, Qwen-VL-Max + SCK + SCRS + RC and GPT4 + SCK + SCRS + RC. The results of these experiments on three distinct datasets are presented in Table V. The outcomes demonstrate a significant improvement in performance for all three baseline models when comparing experiment setting 4 to experiment setting 3. Notably, the GPT4 model exhibited the most pronounced enhancement after the introduction of representative cases. These experimental findings suggest that the inclusion of representative cases is beneficial for improving the capability of model to process multimodal metaphorical information. Through an analysis of these results, we can conclude that exploring the patterns of metaphorical expression within representative cases is critical for guiding MLLM to accurately comprehend metaphors mapped through multimodal information associations.

G. Computational Complexity and Inference Time

1) *Baseline Model GPT4:* As shown in Table VI, during the inference process on the test set, for the task of detecting and interpreting meme images, when only the baseline model GPT4 is adopted, the average response time per image is approximately 3.0 seconds.

TABLE VII

CASE STUDY. ON THIS TABLE, RED WORDS REPRESENT THAT THE REASON IS BOTH REASONABLE AND ACCURATE, GREEN WORDS REPRESENT THAT THE REASON IS REASONABLE BUT NOT ACCURATE ENOUGH

	Case I	Case II	Case III
Hateful Meme			
T5	Mocking your choose by suggesting that are mental illness	Disrespects the dog by likening dog to a group of mem	Disrespects the blacks by suggesting that they are whale shit
GPT4	The hateful meme mock mental illness, thus stigmatizing those with mental health issues.	The hateful meme has a hateful metaphor because it uses dogs to represent people , which can be seen as derogatory and dehumanizing.	This hateful meme exploits racist tropes and associates him with a negative and derogatory stereotype
GPT4 + Ours	This meme disrespects the LGBT community by likening each gender to a mental illness, perpetuating harmful stereotypes and discrimination	This meme disrespects the man by likening them to dogs, implying hostility towards men	This meme disrespects the blacks by comparing them to whale shit , implying that the sharks will not attack them, and they are not worth being treated fairly
Ground Truth	Disrespects the LGBT community by suggesting that each gender is a mental illness	Dehumanizes the males as animals by likening them to dogs	Disrespects the blacks by likening them to whale shit and for that reason why the sharks will not attack them

2) *GPT4+SCK*: After introducing the multimodal generative search algorithm to obtain Socio-Cultural Knowledge (SCK), the average inference time per image extends to 6.71 seconds. Through analysis, the main time consumption stems from the invocation of gpt-3.5-turbo and the google web search api for search and summary generation. This part has an average time consumption of approximately 3.70 seconds per image.

3) *GPT4+SCK+SCRS*: On the basis of the above-mentioned settings, when the Social-Cultural Relevance Scores (SCRS) is further introduced, the average inference time increases to approximately 7.94 seconds. The increase in time cost is mainly attributed to the invocation of gpt-3.5-turbo to perform the relevance evaluation, and its average time cost is approximately 1.23 seconds.

4) *GPT4+SCK+SCRS+RC*: After introducing representative cases (RC) on the basis of Setting 3), the average inference time is approximately 8.05 seconds, and the increase is relatively small. The reasons are as follows: first, the case retrieval process can be parallelly inferred with other steps; second, the faiss vector database used for retrieval has an average retrieval time of less than 100 milliseconds.

H. Case Study

Besides performing quantitative evaluations on the hateful meme reason generation task, we also perform case study to analyze the hateful meme reasons generated by our proposed method, the best-performing baseline model, and the model fine-tuned with supervision.

1) *Case I*: We analyze a hateful meme that combines visual object “a thumbs-up in the colors of the sexual and

gender minorities flag” and textual information “when you get to choose your own mental illness” to metaphorically convey the disrespect towards the sexual and gender minorities community. As shown in Table VII, both VL-T5 and the baseline model GPT4 fail to accurately capture the hateful metaphor of the meme. Specifically, the reason generated by VL-T5 exhibits logical flaws, which fails to delve into the metaphorical meaning of the meme, instead merely reiterating textual information. In contrast, while GPT4 shows improvement in understanding the meme, it erroneously interprets the hateful metaphor of the meme as stigmatization of individuals with mental illnesses.

To overcome these limitations, our approach introduces socio-cultural knowledge: “in the sexual and gender minorities community, people have the freedom to choose their gender identity.” Guided by this socio-cultural knowledge, our model successfully link the term “choose” to “gender”, thereby understanding that the meme disrespect toward the sexual and gender minorities community by likening each gender to a mental illness. This approach based on socio-cultural knowledge significantly enhances the metaphor understanding ability of model.

2) *Case II*: This case that combines the visual element “a group of dogs” with the textual content “a group of men.” to convey gender discrimination against men by likening them to dogs. As shown in Table VII, we observe that the VL-T5 model generates a error metaphorical reason for the meme, with the provided reason being contradictory to the ground truth. Although GPT4 could discern that the meme intends to mock humans by likening them to dogs, it fails to consider the socio-cultural relevance between the visual objects and

TABLE VIII

QUALITATIVE EXPERIMENTS FOR REAL-WORLD HATEFUL MEME. ON THIS TABLE, RED WORDS REPRESENT THAT THE REASON IS BOTH REASONABLE AND ACCURATE, GREEN WORDS REPRESENT THAT THE REASON IS REASONABLE BUT NOT ACCURATE ENOUGH

	Real-world meme I	Real-world meme II	Real-world meme III
Hateful Meme			
URL	URL of real-world meme I	URL of real-world meme II	URL of real-world meme III
T5	Vilifies the Muslims by suggesting that they are terrorists	Vilifies the white by suggesting that they are killing fiend	Disrespects the blacks by suggesting that they are GET OUT
GPT4	The word "halal" is related to Islamic - compliant food and religious and cultural concepts in Islam. Using such language in a derogatory or mocking way is disrespectful to the religious beliefs and cultures of Muslims,	This meme, in a humorous and satirical way, links the scene of Thanos' snap - finger gesture in the Marvel universe to Elon Musk's acquisition of Twitter, implying that his decisions may have a huge impact on Twitter.	This meme with the wide-eyed expression and the text "GET OUT" placed side by side is likely to convey a sense of shock or fear .
GPT4 + Ours	This meme disrespects Muslims by equates the image of Muslims carrying explosives with terrorists , and when combined with the statement like "I heard you are not halal?", it conveys a sense of threat.	This meme satirizes Elon Musk's staff layoff behavior, suggesting that it is just like Thanos' actions and will result in the dismissal of half of Twitter's employees.	This meme expresses racial discrimination against black people by juxtaposing the image of a black person with the text "GET OUT" .

the text from a gender perspective. Consequently, GPT4 did not correctly identify the hateful metaphor is gender discrimination against men.

To address these limitations, our approach assesses the socio-cultural relevance between the visual objects and the textual information from the gender perspective, thereby understanding that the meme disrespects men by likening them to dogs, indicating hostility towards men. In summary, this case study demonstrates the importance of assessing the socio-cultural relevance between visual objects and textual information when analyzing metaphorical reason for the meme.

3) *Case III*: This case combines visual object "black boy" and textual information "why don't sharks attack him? they think it's whale shit," thereby disrespects the blacks by likening them to whale shit and for that reason why the sharks will not attack them. As shown in Table VII, both our method and the baseline model accurately recognize the disrespectful metaphor for blacks. However, compare to the baseline model, our method introduce representative cases to guide the model in producing outputs that are more stylistically and content-wise aligned with the ground truth, effectively enhancing the performance of model.

V. QUALITATIVE EXPERIMENTS FOR REAL-WORLD HATEFUL MEME

In real-world scenarios, the detection of hateful memes often encounters the challenge of ambiguous and dynamically changing content. To verify the practical applicability of our framework, we collect 50 real-world meme images from

the internet for qualitative experimentation. In TABLE VIII, we present the qualitative experimental results of three randomly selected samples. Below is our detailed analysis of the qualitative experimental results:

A. Real-World Meme I

As shown in TABLE VIII, Real-world meme I presents a cartoon character adorned with Muslim decorations carrying a bomb, accompanied by the text "Haven't heard you're not halal?" to convey a sense of threat. This combination uses dual visual and textual implications to metaphorically associate Muslims with terrorists, satirizing the Muslim community as authoritarian and non-inclusive. Our analytical framework successfully explains this complex metaphorical mechanism. Notably, the T5 model, due to the presence of numerous similar cases in its training dataset, is capable of learning this pattern and providing relatively accurate interpretations, though its language expression still lacks fluency and richness. In contrast, GPT-4 fails to recognize the specific metaphorical symbol of "Muslim carrying a bomb," although it speculates that the content may involve satire against Muslims, it does not fully capture the deeper meaning.

B. Real-World Meme II

As shown in TABLE VIII, Real-world meme II combines Elon Musk's image with that of Thanos from the movie *Avengers*, accompanied by the text "kill one half of" and a visual of Twitter employees fading away. This composition

satirizes Musk's large-scale layoffs at Twitter (approximately 50% of employees). Analysis of this case reveals that GPT-4's interpretation is biased due to a lack of relevant sociocultural background knowledge, specifically the fact that "Musk laid off 50% of Twitter employees." On the other hand, the T5 model outputs unreasonable results due to insufficient similar examples in its training data. By contrast, our method leverages internet searches to acquire relevant sociocultural knowledge, enabling precise interpretation of this metaphor.

C. Real-World Meme III

As shown in TABLE VIII, Real-world meme III juxtaposes an image of a Black person with the text "GET OUT," explicitly conveying racial discrimination and hatred toward Black individuals. In this context, GPT-4 misinterprets the core intent of this metaphor due to insufficient consideration of the sociocultural significance embedded in the Black figure's representation. However, our method successfully identifies the underlying racist tendency by focusing on the Black racial identity of the depicted individual, thereby providing a more accurate explanation.

VI. CONCLUSION

In this paper, we analyze the definition of metaphor to discover three key factors of metaphor formation: socio-cultural knowledge, metaphorical tenor, and metaphorical representation pattern. Based on these key factors, we propose a few-shot in-context learning hateful meme detection and interpretation framework. Our framework extracts these key factors from memes and employs them to guide a multimodal large language model to understand the metaphors expressed within memes step-by-step. Experimental results demonstrate that our approach outperforms fully supervised methods, even without pretraining or fine-tuning on manually annotated datasets. Finally, we validate the effectiveness of these three key factors in enhancing model performance on hateful meme detection and reason generation tasks through ablation study and case study.

VII. LIMITATIONS AND FUTURE WORKS

Limitation1: Our framework relies on the LLM's ability to understand socio-cultural knowledge. However, due to potential misalignment between LLMs and humans in cultural cognition, the model may retrieve incorrect knowledge or overlook key information. This phenomenon reduces the accuracy of knowledge retrieval, thereby introducing errors or biases into the metaphor interpretation process.

Future work1: 1) Training a model to dynamically calibrate the accuracy and relevance of the retrieved knowledge. 2) Developing novel alignment techniques to ensure that the LLM's understanding of socio-cultural knowledge aligns more closely with human cognition.

Limitation2: The knowledge base of our framework mainly relies on datasets like FHM and MAMI, which are primarily based on Western culture. Additionally, the training corpus for large language models is predominantly in English and

Chinese, which leads to poor adaptability when the model processes metaphors in the context of niche cultures.

Future work2: 1) Expand the coverage of the visual knowledge base by incorporating more datasets from diverse cultural backgrounds, especially those niche cultures that are currently underrepresented. 2) Develop continuous learning and adaptive mechanisms that enable the model to continuously learn and optimize its understanding of niche cultures over time.

Limitation3: The complexity of our framework may lead to higher inference times, posing challenges for real-time applications or integration into existing content moderation systems. Therefore, it is necessary to optimize the proposed method through engineering adjustments according to specific application scenarios.

Future work3: To enhance real-time performance, we implemented the following optimizations in practical application scenarios: 1) By constructing a domain-specific socio-cultural knowledge base with strong generalization capabilities tailored to actual application needs, we eliminated the online search component. 2) We deployed smaller-scale large language models (e.g., 2B or 7B parameters) locally using the high-performance parallel processing framework MLLM, significantly improving model response time.

REFERENCES

- [1] S. M. Anurudu and I. M. Obi, "Decoding the metaphor of internet meme: A study of satirical tweets on black friday sales in Nigeria," *AFRREV LALIGENS, Int. J. Lang., Literature Gender Stud.*, vol. 6, no. 1, pp. 91–100, Apr. 2017.
- [2] M. Mitsiaki, "Investigating metaphor in modern Greek internet MEMES: An applied approach with L2 pedagogical implications," *Revista Brasileira De Alfabetização*, no. 12, pp. 73–106, Jul. 2020.
- [3] D. Liu, P. Zhou, Z. Xu, H. Wang, and R. Li, "Few-shot temporal sentence grounding via memory-guided semantic learning," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 33, no. 5, pp. 2491–2505, May 2023.
- [4] J. Shen, D. Tao, and X. Li, "Modality mixture projections for semantic video event detection," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 18, no. 11, pp. 1587–1596, Nov. 2008.
- [5] M. Jin, W. Hu, L. Zhu, X. Wang, and R. Hong, "Based on spatial and temporal implicit semantic relational inference for cross-modal retrieval," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 11, pp. 11286–11298, Nov. 2024.
- [6] P. Lippe et al., "A multimodal framework for the detection of hateful memes," 2020, *arXiv:2012.12871*.
- [7] N. Muennighoff, "Vilio: State-of-the-art visio-linguistic models applied to hateful memes," 2020, *arXiv:2012.07788*.
- [8] R. Cao, R. Ka-Wei Lee, W.-H. Chong, and J. Jiang, "Prompting for multimodal hateful meme classification," 2023, *arXiv:2302.04156*.
- [9] S. Sharma, M. Shad Akhtar, P. Nakov, and T. Chakraborty, "DISARM: Detecting the victims targeted by harmful memes," 2022, *arXiv:2205.05738*.
- [10] M. Shan Hee, W.-H. Chong, and R. Ka-Wei Lee, "Decoding the underlying meaning of multimodal hateful memes," 2023, *arXiv:2305.17678*.
- [11] D. Kiela et al., "The hateful memes challenge: Detecting hate speech in multimodal memes," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 33, 2020, pp. 2611–2624.
- [12] G. Campbell, *The Philosophy of Rhetoric*. Carbondale, IL, USA: SIU Press, 1988.
- [13] J. Weiner, "Oxford English dictionary," Simpson, Oxford Univ. Press, New York, NY, USA, Tech. Rep., 1989, doi: [10.1136/bmj.1.3503.311-a](https://doi.org/10.1136/bmj.1.3503.311-a).
- [14] R. Vacca et al., "What do you meme? Students communicating their experiences, intuitions, and biases surrounding data through memes," in *Proc. 21st Annu. ACM Interact. Design Children Conf.*, 2022, pp. 212–224.
- [15] N. Zhang, "Toward a phenomenological understanding of internet-mediated meme-ing as a lived experience in social distancing via autoethnography," *J. Contemp. Ethnography*, vol. 53, no. 2, pp. 181–211, Apr. 2024.

- [16] J. Liu, M. Yang, C. Li, and R. Xu, "Improving cross-modal image-text retrieval with Teacher-Student learning," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 31, no. 8, pp. 3242–3253, Aug. 2021.
- [17] Z. Liu, F. Chen, J. Xu, W. Pei, and G. Lu, "Image-text retrieval with cross-modal semantic importance consistency," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 33, no. 5, pp. 2465–2476, May 2023.
- [18] M. Cao, Y. Bai, Z. Cao, L. Nie, and M. Zhang, "Efficient image-text retrieval via keyword-guided pre-screening," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 6, pp. 5132–5145, Jun. 2024.
- [19] W. Li, R. Xiong, and X. Fan, "Multi-layer probabilistic association reasoning network for image-text retrieval," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 10, pp. 9706–9717, Oct. 2024.
- [20] J. Li, W. K. Wong, L. Jiang, X. Fang, S. Xie, and Y. Xu, "CKDH: CLIP-based knowledge distillation hashing for cross-modal retrieval," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 7, pp. 6530–6541, Jul. 2024.
- [21] A. Radford et al., "Learning transferable visual models from natural language supervision," in *Proc. ICML*, 2021, pp. 8748–8763.
- [22] S. Suryawanshi, B. R. Chakravarthi, M. Arcan, and P. Buitelaar, "Multimodal meme dataset (MultiOFF) for identifying offensive content in image and text," in *Proc. 2nd Workshop Trolling, Aggression Cyberbullying*, 2020, pp. 32–41.
- [23] S. Pramanick et al., "Detecting harmful memes and their targets," 2021, *arXiv:2110.00413*.
- [24] C. Yang, F. Zhu, G. Liu, J. Han, and S. Hu, "Multimodal hate speech detection via cross-domain knowledge transfer," in *Proc. 30th ACM Int. Conf. Multimedia*, 2022, pp. 4505–4514.
- [25] M. S. Hee, R. K.-W. Lee, and W.-H. Chong, "On explaining multimodal hateful meme detection models," in *Proc. ACM Web Conf.*, Apr. 2022, pp. 3651–3655.
- [26] Z. Zhao et al., "Recommender systems in the era of large language models (LLMs)," *IEEE Trans. Knowl. Data Eng.*, vol. 36, no. 11, pp. 6889–6907, Nov. 2024.
- [27] J. Li et al., "Empowering molecule discovery for molecule-caption translation with large language models: A ChatGPT perspective," *IEEE Trans. Knowl. Data Eng.*, vol. 36, no. 11, pp. 6071–6083, Nov. 2024.
- [28] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, Jun. 2017, pp. 5998–6008.
- [29] T. B. Brown et al., "Language models are few-shot learners," in *Proc. Adv. Neural Inf. Process. Syst.*, 2020, pp. 1877–1901.
- [30] A. Chowdhery et al., "PaLM: Scaling language modeling with pathways," *J. Mach. Learn. Res.*, vol. 24, no. 240, pp. 1–113, 2023.
- [31] H. Touvron et al., "Llama 2: Open foundation and fine-tuned chat models," 2023, *arXiv:2307.09288*.
- [32] OpenAI et al., "GPT-4 technical report," 2023, *arXiv:2303.08774*.
- [33] H. Liu, C. Li, Q. Wu, and Y. J. Lee, "Visual instruction tuning," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 36, 2023, pp. 34892–34916.
- [34] D. Zhu, J. Chen, X. Shen, X. Li, and M. Elhoseiny, "MiniGPT-4: Enhancing vision-language understanding with advanced large language models," 2023, *arXiv:2304.10592*.
- [35] G. Team et al., "Gemini: A family of highly capable multimodal models," 2023, *arXiv:2312.11805*.
- [36] J. Li, D. Li, S. Savarese, and S. Hoi, "Blip-2: Bootstrapping language-image pre-training with frozen image encoders and large language models," in *Proc. Int. Conf. Mach. Learn.*, 2023, pp. 19730–19742.
- [37] A. Yang et al., "Qwen2 technical report," 2024, *arXiv:2407.10671*.
- [38] T. Cheng, L. Song, Y. Ge, W. Liu, X. Wang, and Y. Shan, "YOLO-world: Real-time open-vocabulary object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 16901–16911.
- [39] S. Pramanick, S. Sharma, D. Dimitrov, M. Shad Akhtar, P. Nakov, and T. Chakraborty, "MOMENTA: A multimodal framework for detecting harmful memes and their targets," 2021, *arXiv:2109.05184*.
- [40] E. Fersini et al., "SemEval-2022 task 5: Multimedia automatic misogyny identification," in *Proc. 16th Int. Workshop Semantic Eval. (SemEval-)*, Jan. 2022, pp. 533–549.
- [41] Y. Wang, R. Sun, N. Luo, Y. Pan, and T. Zhang, "Image-to-image matching via foundation models: A new perspective for open-vocabulary semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 3952–3963.
- [42] M. Oquab et al., "DINOv2: Learning robust visual features without supervision," 2023, *arXiv:2304.07193*.
- [43] L. Mathias et al., "Findings of the WOA5 shared task on fine grained hateful memes detection," in *Proc. 5th Workshop Online Abuse Harms (WOAH)*, 2021, pp. 201–206.
- [44] W. Xiang, "Analysis of the time complexity of quick sort algorithm," in *Proc. Int. Conf. Inf. Manage., Innov. Manage. Ind. Eng.*, vol. 1, Nov. 2011, pp. 408–410.
- [45] W. Fan et al., "A survey on RAG meeting LLMs: Towards retrieval-augmented large language models," 2024, *arXiv:2405.06211*.
- [46] P. C.-H. Lam, L. Chu, M. Torgonskiy, J. Pei, Y. Zhang, and L. Wang, "Finding representative interpretations on convolutional neural networks," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2021, pp. 1345–1354.
- [47] A. Nayak and A. Agrawal, "Detection of hate speech in social media memes: A comparative analysis," in *Proc. 3rd Int. Conf. Intell. Comput. Instrum. Control Technol. (ICICICT)*, Aug. 2022, pp. 1179–1185.
- [48] J. Wei et al., "Chain-of-thought prompting elicits reasoning in large language models," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 35, 2022, pp. 24824–24837.
- [49] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," 2018, *arXiv:1810.04805*.
- [50] T.-Y. Lin et al., "Microsoft COCO: Common objects in context," in *Proc. ECCV*, vol. 14, Zurich, Switzerland. Cham, Switzerland: Springer, Sep. 2014, pp. 740–755.
- [51] P. Sharma, N. Ding, S. Goodman, and R. Soicrut, "Conceptual captions: A cleaned, hypemymed, image alt-text dataset for automatic image captioning," in *Proc. 56th Annu. Meeting Assoc. Comput. Linguistics*, 2018, pp. 2556–2565.
- [52] D. Kiela, S. Bhooshan, H. Firooz, E. Perez, and D. Testuggine, "Supervised multimodal bitransformers for classifying images and text," 2019, *arXiv:1909.02950*.
- [53] J. Li, D. Li, C. Xiong, and S. Hoi, "BLIP: Bootstrapping language-image pre-training for unified vision-language understanding and generation," in *Proc. ICML*, 2022, pp. 12888–12900.
- [54] J. Li, R. Selvaraju, A. Gotmare, S. Joty, C. Xiong, and S. C. H. Hoi, "Align before fuse: Vision and language representation learning with momentum distillation," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 34, 2021, pp. 9694–9705.
- [55] R. Cao, M. S. Hee, A. Kuek, W.-H. Chong, R. K.-W. Lee, and J. Jiang, "Pro-cap: Leveraging a frozen vision-language model for hateful meme detection," in *Proc. 31st ACM Int. Conf. Multimedia*, pp. 5244–5252, Oct. 2023.
- [56] J. Mei, J. Chen, W. Lin, B. Byrne, and M. Tomalin, "Improving hateful meme detection through retrieval-guided contrastive learning," in *Proc. 62nd Annu. Meeting Assoc. Comput. Linguistics Long Papers*, vol. 1, 2024, pp. 5333–5347.
- [57] J. Cho, J. Lei, H. Tan, and M. Bansal, "Unifying vision-and-language tasks via text generation," in *Proc. ICML*, 2021, pp. 1931–1942.
- [58] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2016, pp. 770–778.
- [59] J. M. Lobo, A. Jiménez-Valverde, and R. Real, "AUC: A misleading measure of the performance of predictive distribution models," *Global Ecol. Biogeography*, vol. 17, no. 2, pp. 145–151, Mar. 2008.
- [60] K. Papineni, S. Roukos, T. Ward, and W.-J. Zhu, "Bleu: A method for automatic evaluation of machine translation," in *Proc. 40th Annu. Meeting Assoc. Comput. Linguistics*, 2002, pp. 311–318.
- [61] C.-Y. Lin, "Rouge: A package for automatic evaluation of summaries," in *Proc. Text Summarization Branches Out*, 2004, pp. 74–81.
- [62] A. Joshi, S. Kale, S. Chandel, and D. Pal, "Likert scale: Explored and explained," *Brit. J. Appl. Sci. Technol.*, vol. 7, no. 4, pp. 396–403, Jan. 2015.



Xiuxian Wang is currently pursuing the Ph.D. degree with the School of Electronic and Information Engineering, Tianjin University, Tianjin, China. His research interests include computer vision and machine learning.



Lanjun Wang (Member, IEEE) received the Ph.D. degree in electronic engineering from Tsinghua University, Beijing, China, in 2011. She was an Exchange Student with California Institute of Advanced Transportation Technology, University of California at Berkeley, USA, and a Post-Doctoral Researcher with the University of Waterloo, Canada. She is currently a Professor with the School of New Media and Communication, Tianjin University. Her current research interests include interpretability, robustness, fairness, transparency, and other new technologies that ensure the safe development of artificial intelligence systems. At present, the main focus is on the cross-cutting issues related to intelligent communication under generative models.



Guoqing Jin is currently with the State Key Laboratory of Communication Content Cognition, People's Daily Online, Beijing, China.



Yuting Su (Member, IEEE) was a Visiting Professor with Case Western Reserve University, USA. He is currently a Professor with the School of Electrical and Information Engineering, Tianjin University. His current research interests include computer vision and machine learning.



Hongshuo Tian is currently a Post-Doctoral Researcher with the School of Electronic and Information Engineering, Tianjin University, China. His research interests include computer vision and machine learning.



An-An Liu (Senior Member, IEEE) received the Ph.D. degree in electronic engineering from Tianjin University, Tianjin, China, in 2010. He was a Visiting Professor with the SeSaMe Centre, National University of Singapore, Singapore, and a Visiting Scholar with the Robotics Institute, Carnegie Mellon University, Pittsburgh, PA, USA. He is currently a Professor with the School of Electrical and Information Engineering, Tianjin University. His current research interests include computer vision and machine learning.